Foreign Exchange Market and Microstructural Approach to Exchange Rate Dynamics

1MT404 Financial Stability

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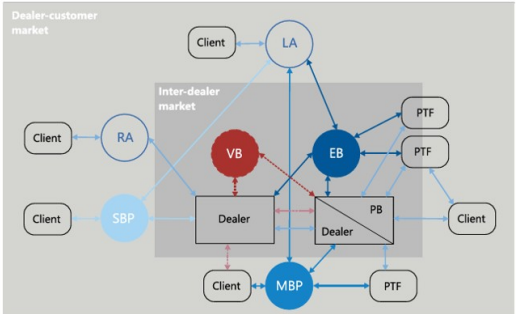
October 28, 2026

Presentation Content

1. Structure and trends of the foreign exchange market
2. Carry trades – theory, empirics, and risks
3. Role of FX Hedging in Exchange Rate Dynamics
4. Algorithmic and High-Frequency Trading (HFT)
5. Hedge Funds as a Key Client in the FX Market
6. Standard macroeconomic approach and its limits
7. Microstructural Approach – Evans-Lyons Model
8. FX and the Global Financial Cycle
9. Conclusion and Implications for Financial Stability

Fragmented Structure of the FX Market

- **Decentralized OTC Market** (Over-the-Counter) – no central exchange exists.
- **Three Main Segments**:
  - Dealer-to-Dealer (D2D)
  - Dealer-to-Client (D2C)
  - Client-to-Client (C2C)
- **BIS Triennial Survey 2025**: Average daily turnover amounts to **$9.6 trillion** (+28% compared to April 2022).
- Spot 31%, FX swaps 42%, outright forwards 19%, options 7%; USD on one side of 89% of all trades



EB = electronic broker; LA = liquidity aggregator; MBP = multi-bank platform; PB = prime broker; PTF = principal trading firm; RA = retail aggregator; SBP = single-bank platform; VB = voice broker. Dashed lines indicate voice execution; solid lines indicate electronic execution.

Source: Schrimpf, A. and Sushko, V. (2019), "FX trade execution: complex and highly fragmented", BIS Quarterly Review, December, Graph 1.

Trend Changes in Structure by Counterparty

- Changes in FX market structure during overall turnover growth (+28% 2022–2025), with short-term liquidity drops during crises/shocks
- **Inter-dealer share**: from roughly 60% (2001) to below 40% (2019), back to 46% in 2025 – more client flow is **internalised** by a few large dealers instead of being recycled in the inter-dealer market (concentration of market making)
- **Other financial institutions (OFIs)**: 50% of turnover in 2025 (47% in 2022)
  - non-reporting (smaller) banks 24%, institutional investors 13%, hedge funds and proprietary trading firms (PTFs) 8%
  - roughly half of the "OFI" segment are still banks – not all of it is "non-bank"
- **Non-financial customers**: only 5% of turnover (7% in 2019)
- Concentration: Top 10 liquidity providers $\approx$ 65% of volume (Euromoney survey 2019, down from $\approx$80% in 2007–2014)

Why Has the "Other Financial Institutions" Segment Grown?

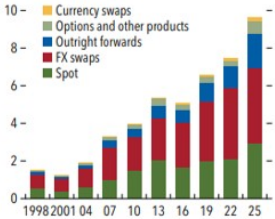
- Algorithmic/high-frequency trading
- Shift of small banks into the client sector
- Development of professional asset managers (hedge funds, mutual, pension, and investment funds)
- Higher international diversification of portfolio investors, including subsequent hedging of exchange rate risks (response to asset growth of non-bank intermediaries)
- Development of leveraged investors (hedge funds, retail investors) with very short investment horizons (margin trading, credit-based, carry trade, complex investment strategies)
- Decline in corporate client trading, influence of centralized cash management in multinational corporations
- Growth of retail investor trading activity via online platforms (retail aggregators - eToro, Plus500, IG)

Trading Volume and FX Market Structure by Counterparty 1

Figure 2.1. Key Developments in the Global Foreign Exchange Market

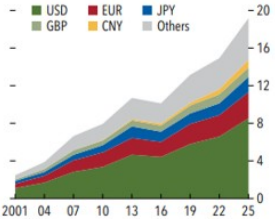
The FX market has grown substantially, driven mainly by an increase in swap trades.

1. FX Market Turnover, by Instrument
(Daily average trillions of dollars, net-net basis)



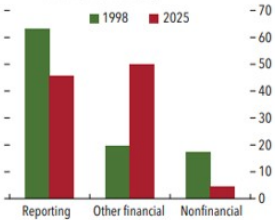
The US dollar continues to be the dominant currency in global FX trading.

2. FX Market Turnover, by Currency
(Daily average trillions of dollars, net-net basis)



The shares of participants have shifted, with nonbank financial institutions taking a greater role.

3. FX Market Turnover, by Counterparty
(Daily average percent of total market turnover, net-net basis)

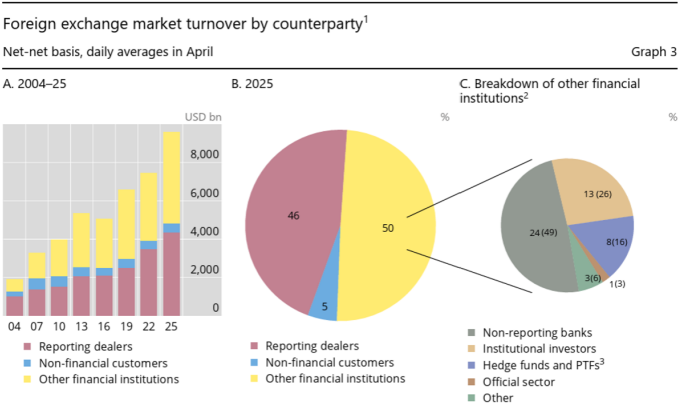


Sources: BIS 2025a; and IMF staff calculations.

Note: Daily averages correspond to April. The BIS Triennial Central Bank Survey (2025a) adjusts figures reported on a net-net basis to correct for double counting in local and cross-border interdealer transactions. In panel 2, because each foreign exchange transaction involves two currencies, the sum of the individual currencies is twice that of the total turnover involved. CNY = Chinese yuan; EUR = euro; FX = foreign exchange; GBP = British pound; JPY = Japanese yen; USD = US dollar.

Source: IMF (2025), Global Financial Stability Report, October, Chapter 2 "Risk and Resilience in the Global Foreign Exchange Market".

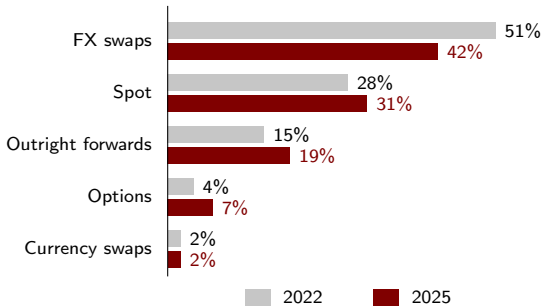
Trading Volume and FX Market Structure by Counterparty 2



Source: BIS (2025), Triennial Central Bank Survey: OTC foreign exchange turnover in April 2025.

Derivative Trades in Detail

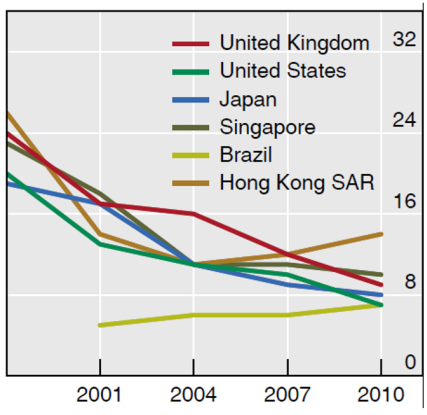
Global FX turnover by instrument (share of total, April)

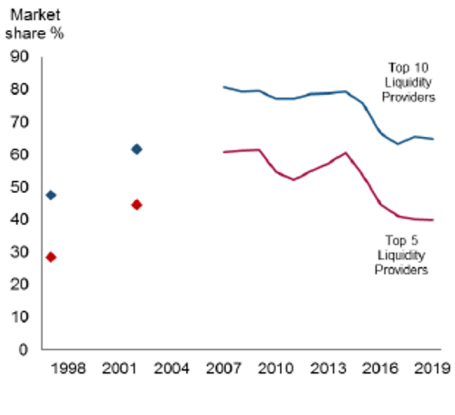


- Derivatives (swaps, forwards, options, currency swaps) $\approx$ 70% of turnover
- Growth 2022–2025: spot $+42\%$, outright forwards $+60\%$, FX swaps only $+5\%$
- Options' share almost doubled (4% $\rightarrow$ 7%) – more demand for convexity and hedging of tail risk
- FX swaps remain the largest instrument (USD 4 trillion a day) – the funding leg of hedging and of the global dollar system

Source: BIS Triennial Central Bank Survey 2025.

FX Market Concentration

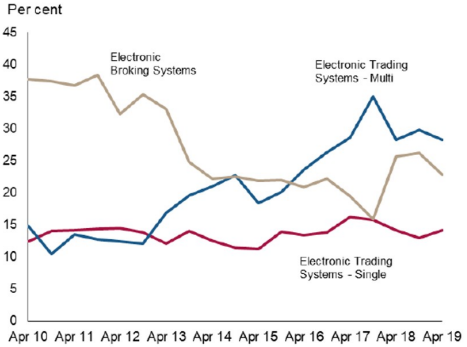




Source: Right panel: Euromoney FX surveys.

Development of Electronic Trading Systems

- Development of e-brokers and multi-bank trading systems
- Creation of trading platforms for algorithmic/high-frequency trading
- Development of prime-brokerage and web-based trading systems for retail clients



Source: London Foreign Exchange Joint Standing Committee surveys and Bank of England calculations.

Source: London Foreign Exchange Joint Standing Committee turnover surveys; Bank of England calculations.

Carry Trades – Definition and Motivation

Carry trade = speculative leveraged strategy exploiting the violation of **Uncovered Interest Parity (UIP)**

$$\begin{aligned}\text{Carry return} &= (i^* - i) + \Delta s_{t+1} \\ &\approx \text{interest differential} + \text{change in spot rate}\end{aligned}$$

(s = log price of the foreign/target currency in units of the funding currency; borrow at i, invest at i^*)
Funding currency (low interest): $\rightarrow$ JPY, CHF, USD (historically)
Target currency (high interest): $\rightarrow$ AUD, NZD, MXN, BRL, TRY, HUF, PLN, USD

Forward Premium Puzzle (Fama 1984)

- UIP Theory: $E_t[\Delta s_{t+1}] = i_t - i_t^*$ $\Rightarrow$ high-interest currency should depreciate (s = log domestic price of foreign currency)
- Fama regression, using covered parity $f_t - s_t \approx i_t - i_t^*$:

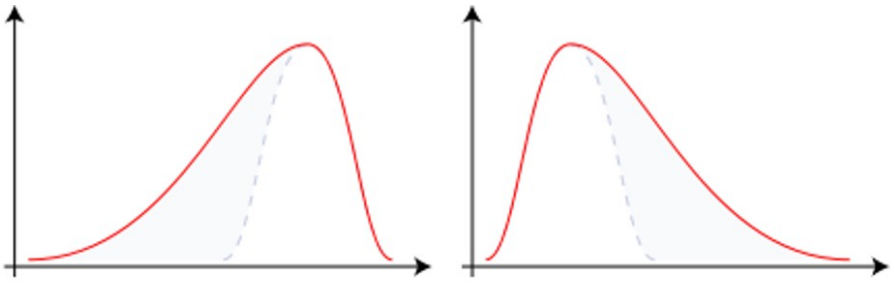
$$\Delta s_{t+1} = \alpha + \beta\,(f_t - s_t) + \varepsilon_{t+1}, \qquad \text{UIP + rational expectations} \Rightarrow \alpha = 0,\ \beta = 1$$

- Empirics: $\hat{\beta}$ well below 1 and often negative (≈ -0.5 to -3) $\Rightarrow$ high-interest currencies do not depreciate enough and frequently **appreciate** $\rightarrow$ positive average carry return
- Carry trade return = **risk premium** for *crash risk* (and global FX volatility risk), possibly plus expectational errors

Crash Risk in Carry Trades – Brunnermeier et al. (2008, 2009)

- Investment currencies exhibit strong **negative skewness**
- Carry trade unwinds occur during:
  - Growth in global risk aversion (VIX spike)
  - Deterioration of funding liquidity
  - Margin calls on leveraged positions
- „Go up by the stairs, down by the elevator“
- Crash insurance becomes expensive as positions build up: the 25-delta risk reversal on the **investment** currency (call − put) becomes more **negative** – OTM puts on AUD, NZD, MXN are bid
- Sign depends on quoting convention: on USD/JPY, a negative risk reversal means yen calls are bid (crash protection for yen shorts)
- Cross-section (see “Skewness and Risk Reversal within Carry Trades”): the higher the interest differential, the more negative both skewness and the risk reversal

Skewness of Probability Distribution

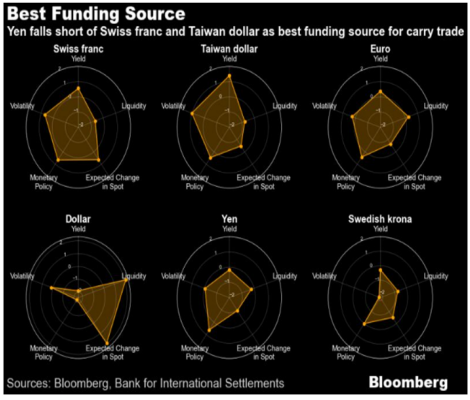


Carry Trade Attractiveness Metrics

Indicator	Formula	Interpretation
Carry-to-risk	$\frac{i^* - i}{\sigma^{imp}}$	ex-ante attractiveness
Sharpe ratio	$\frac{\mathbb{E}[r_{carry}]}{\sigma(r_{carry})}$	ex-post performance
Risk reversal (25-delta)	$IV_{call}^{25\Delta} - IV_{put}^{25\Delta}$	skewness (crash risk)

- **carry to risk** – ex-ante measure of attractiveness as the ratio of interest differential to implied volatility
- **Sharpe ratio** – ex-post measure: average return over realized volatility of the strategy; a funded carry trade is a zero-cost position, so its return is already an excess return (no r_f to subtract)
- **risk reversal** – difference between prices (implied volatilities) of two „out of the money“ call and put options

Conditions for Carry Trade 1





Source: Bloomberg

Source: Bloomberg (2022), "Traders need to rethink tumbling yen as a carry funding favorite"; data: Bloomberg, BIS.

Cross-currency basis as a stress indicator

Covered interest parity, with S and F in units of foreign currency (FC) per USD:

$$(1 + i_{USD}) = \frac{S}{F}(1 + i_{FC})$$

In reality (Du, Tepper & Verdelhan 2018 define the basis x):

$$(1 + i_{USD}) = \frac{S}{F}(1 + i_{FC} + x), \qquad x \neq 0$$

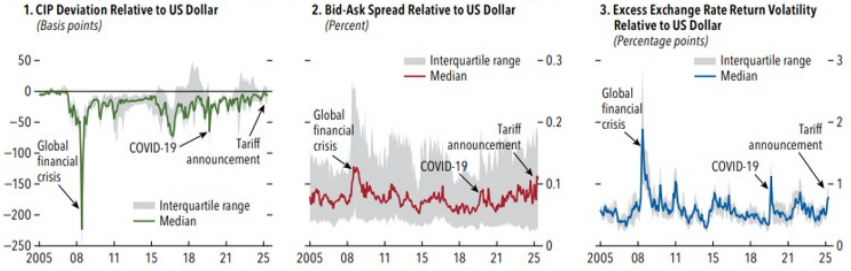
- Persistent since 2008: limits to arbitrage (balance-sheet costs, quarter-ends)
- Negative basis ($x < 0$): dollars obtained synthetically through an FX swap cost more than cash dollars $\Rightarrow$ USD shortage premium
- Sharp movements during crises (2008, 2020).
- Signal of global dollar liquidity.

FX market = global liquidity market, not just exchange rates.

Conditions for Carry Trade 2

Figure 2.2. Macrofinancial Uncertainty and Foreign Exchange Market Conditions

Foreign exchange funding and market liquidity conditions deteriorated significantly during the global financial crisis and the COVID-19 market turmoil.

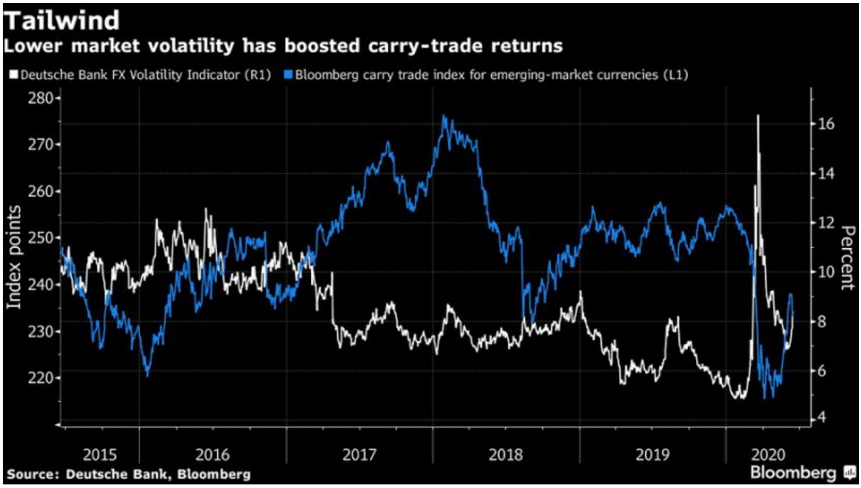


Sources: Bloomberg Finance L.P.; LSEG Datastream; and IMF staff calculations.

Note: In panel 1, CIP deviation is calculated using three-month overnight index swap rates for 12 currencies against the US dollar. A negative widening basis signals stress in dollar funding markets. Panel 2 shows the bid-ask spread calculated as [(ask rate − bid rate)/mid rate] × 100. Wider spreads suggest reduced market liquidity. In panel 3, excess exchange rate return = log(exchange rate at time t/exchange rate at time $t-1$) − log(forward rate at time $t-1$/exchange rate at time $t-1$). In all panels, "tariff announcement" refers to the April 2, 2025, US declaration of new import tariff rates. See Online Annex Figure 2.3.1 for details on the construction of the measures used in this figure and illustrations by country group (that is, advanced economies and emerging market economies). CIP = coverage interest parity.

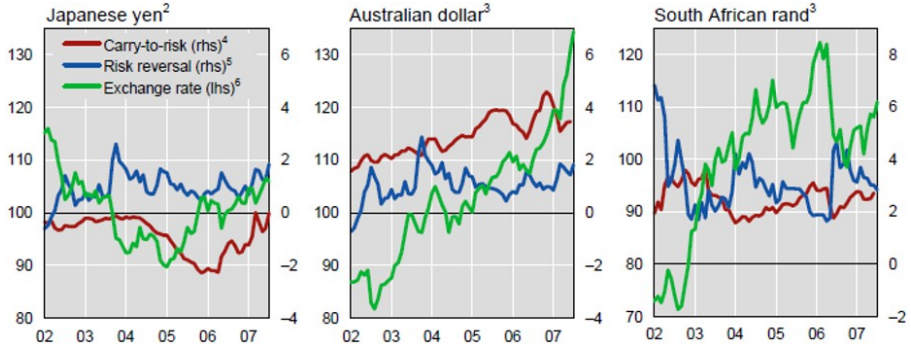
Source: IMF (2025), Global Financial Stability Report, October, Chapter 2 "Risk and Resilience in the Global Foreign Exchange Market", Figure 2.2.

Exchange Rate Volatility and Carry Trade Activity



Source: Bloomberg (2020).

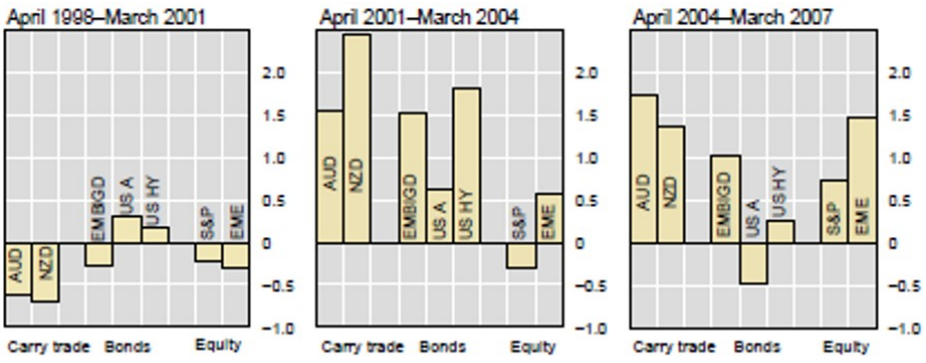
Carry-to-risk Ratio and Risk Reversal



[1] Central bank target interest rates, in per cent; for the Japanese yen, BOJ target rate; for the Swiss franc, Libor target rate; for the US dollar, federal funds rate; for the Australian dollar and the New Zealand dollar, cash rate; for the pound sterling, base rate; for the Brazilian real, Selic rate; for the South African rand, repo rate; for the Mexican peso, official overnight closing bank rate. [2] Against the US dollar. [3] Against the Japanese yen. [4] Defined as the three-month interest rate differential divided by the implied volatility derived from three-month at-the-money exchange rate options; quintuple scale (eg the number 2 represents a ratio of 0.4). [5] A positive value indicates a willingness to pay more to hedge against a yen appreciation. [6] 2002–06 = 100.

Source: Galati, G., Heath, A. and McGuire, P. (2007), "Evidence of carry trade activity", BIS Quarterly Review, September, Graph 1.

Sharpe Ratio for Various Investment Strategies



AUD = Australian dollar; NZD = New Zealand dollar; EMBIGD = JPMorgan Chase EMBI Global Diversified index; US A = Merrill Lynch US dollar A-rated corporate index; US HY = Merrill Lynch US dollar high-yield corporate index; S&P = S&P 500 equity index; EME = MSCI emerging markets equity index.

Consequences of Massive Carry Trades

- Long-term depreciation of funding currency/appreciation of target currency followed by sudden appreciation of funding currency/depreciation of target currency
- Trend reversal is triggered by fundamentally insignificant price-setting information affecting the expected rate, or due to budget constraints on the growth of short positions in the funding currency
- Increase in bid/ask spread and exchange rate volatility at the moment of trend reversal
- Resulting negative skewness of the distribution of exchange rate changes for the investment currency (positive skewness for funding currency) as a result of limits on funding currency volume
- Carry trade strategy return includes a premium for „crash risk“, i.e., the risk of a negative shock in the reversal of funding/investment currency movement

Yen Carry Trade Unwind: August 2024 and After

August 2024 – the sharpest carry unwind in years: Nikkei -12.4% on 5 August, its worst day since 1987

- BoJ exits negative rates (March 2024) and hikes to 0.25% on 31 July 2024
- Weak US payrolls (2 August) $\rightarrow$ expected Fed cuts $\rightarrow$ narrowing rate differential $\rightarrow$ massive JPY short covering
- USD/JPY from $\approx$162 (early July) to $\approx$142 (5 August): JPY appreciation of $\approx$12% in about 3.5 weeks
- Global impact: S&P 500 $\approx -8\%$ from its mid-July peak, VIX above 60 intraday on 5 August, high-yield EM currencies depreciated sharply
- BIS Bulletin No. 90: yen carry before the event $\approx$ ¥40 trillion (USD 250 bn), probably underestimated; the reaction to US data was amplified by deleveraging and margin increases

2025–2026: still a funding currency

- BoJ normalisation continues: 0.5% (January 2025) ... 1.25% (September 2026, highest since 1995) – but the rate gap vs USD and EM remains large $\rightarrow$ new carry trades (MXN/JPY, frontier EM)
- USD/JPY back above 160 at the end of April 2026 $\rightarrow$ Ministry of Finance buys ¥11.7 trillion of yen (28 April – 27 May 2026), the largest intervention on record (case study in the microstructure section)

Lesson

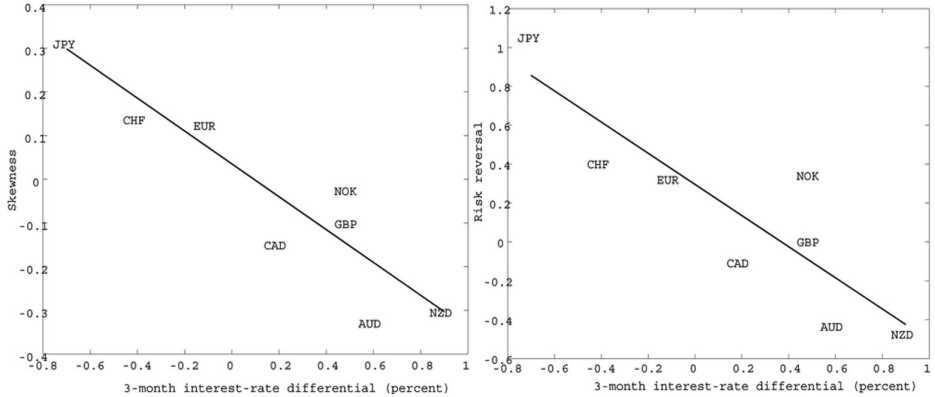
Normalisation in the funding currency is the classic trigger of carry unwinds – but the timing depends on **positioning**, not on the size of the hike (0.25 pp in July 2024 was enough).

Historical Example of Carry Trade Impact on USD/JPY Dynamics



Source: Brunnermeier, M., Nagel, S. and Pedersen, L. (2009), "Carry trades and currency crashes".

Skewness and Risk Reversal within Carry Trades

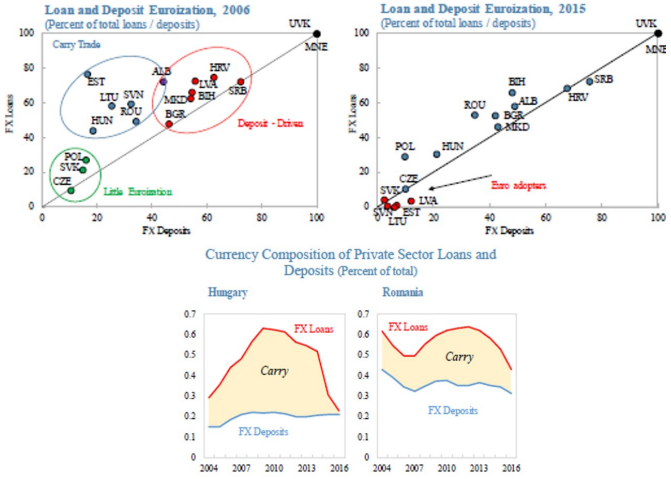


Source: Brunnermeier, M., Nagel, S. and Pedersen, L. (2009), "Carry trades and currency crashes".

Foreign Currency Mortgages as a Special Case of Carry Trades

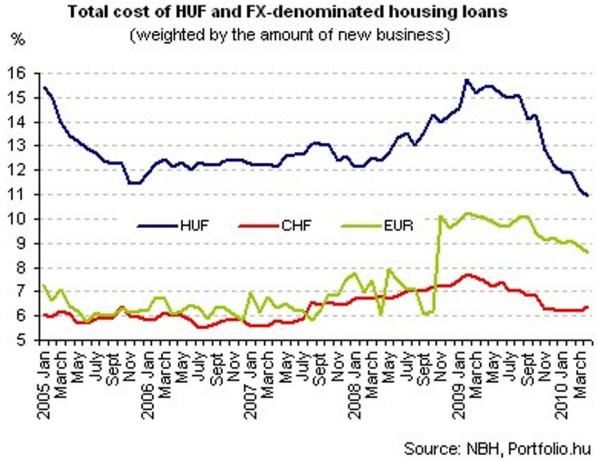
- Long-term positive interest differential between selected transition economies (Poland, Hungary, Croatia) and developed countries
- Long-term trend appreciation of transition economy currencies
- Preference for foreign currency denomination of mortgage loans without hedging currency risk
- Flight to safe assets during crises (German bonds, gold, and CHF)
- CHF appreciation against USD and resulting depreciation of transition currencies against CHF
- Growth in default rates of foreign currency mortgages (transfer of client currency risk to bank credit risk) to approximately double that of domestic currency mortgages
- Decline in real household wealth and consumption
- Pressure for non-market solutions (fixing exchange rates at the expense of banks, early repayment at non-market rates, interest rate subsidies, restrictions on selling foreclosed property, state buyouts)

Foreign Currency Loans and Deposits



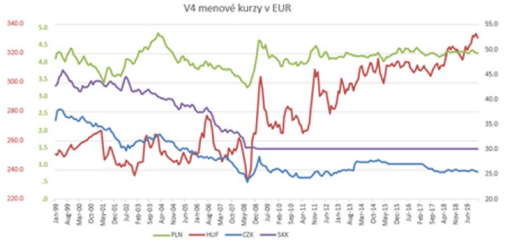
Source: Geng, N., Scutaru, T. and Wiegand, J. (2018), "Carry trade vs. deposit-driven euroization", IMF Working Paper 18/58.

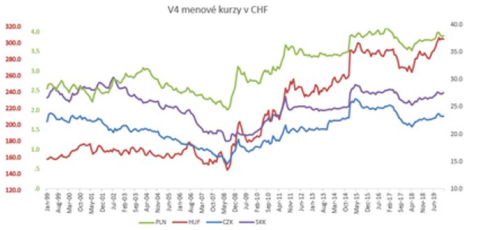
Weighted Mortgage Costs by Currency



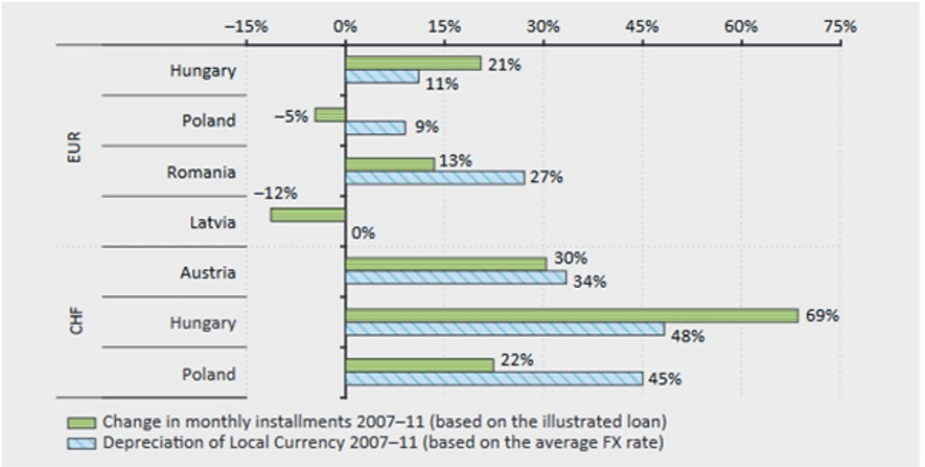
Source: Magyar Nemzeti Bank; Portfolio.hu.

CHF/PLN and CHF/HUF Exchange Rate Evolution



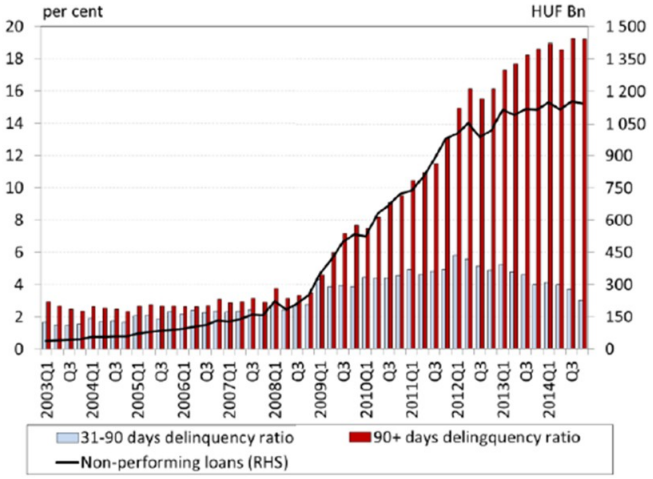


Mortgage Installment Growth in Hungary



Source: Bethlendi (2015); Szigel (2012); Magyar Nemzeti Bank.

NPLs in Hungary



Rising Importance of FX Hedging Post-GFC

- Since the Global Financial Crisis, massive growth in portfolio investments by non-US investors into USD assets
- BIS data 2025: non-US investors hold over **USD 17.6 trillion** in US equities and **USD 13.6 trillion** in US bonds
- Parallel growth in FX derivatives (swaps, forwards, options) – notional value of FX derivatives at end of 2024: **USD 130 trillion**
- Hedge ratio (share of hedged positions) for many investors fell in 2022–2024 due to high hedging costs (high USD rates)

Source: Shin, Wooldridge, Xia (2025) – BIS Bulletin No. 105 (20 June 2025)

Hedging Mechanics and Influence on Spot Rate

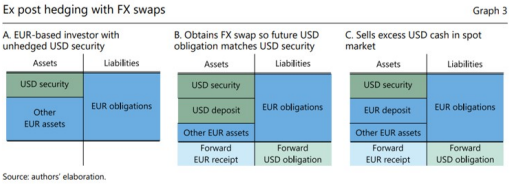
Hedged Purchase (Initial)

- Buy USD asset + simultaneous forward sell USD
- Spot impact ≈ 0 (buy and sell offset each other)

Ex-post Change in Hedge Ratio

- Investor already holds unhedged position and decides to hedge additionally
- Entering FX swap/forward $\rightarrow$ mismatch between spot and forward flows
- Result: significant pressure on spot rate

Example: Euro investor increases hedge ratio $\rightarrow$ sells USD spot $\rightarrow$ USD weakens against EUR



Source: Shin, H. S., Wooldridge, P. and Xia, D. (2025), "US dollar's slide in April 2025: the role of FX hedging", BIS Bulletin No. 105.

Case Study: USD Drop in April 2025

- April 2025: Announcement of new US tariffs → „triple decline“ (stocks, bonds, and dollar fell simultaneously)
- Non-US investors (especially Asian) had low hedge ratios → during USD depreciation, they began increasing hedging ex-post
- Largest dollar weakening occurred during Asian trading hours
- Cross-currency basis worsened significantly (negative) → hedging became more expensive
- Hedging flows amplified USD depreciation, despite no massive sell-off of US bonds

BIS Bulletin 105 (2025): currency hedging by non-US holders of dollar securities „appears to have made an important contribution“ to the weakness of the dollar in April and May 2025.

Case Study: Taiwan Dollar, May 2025 – Hedging Flows in a Thin Market

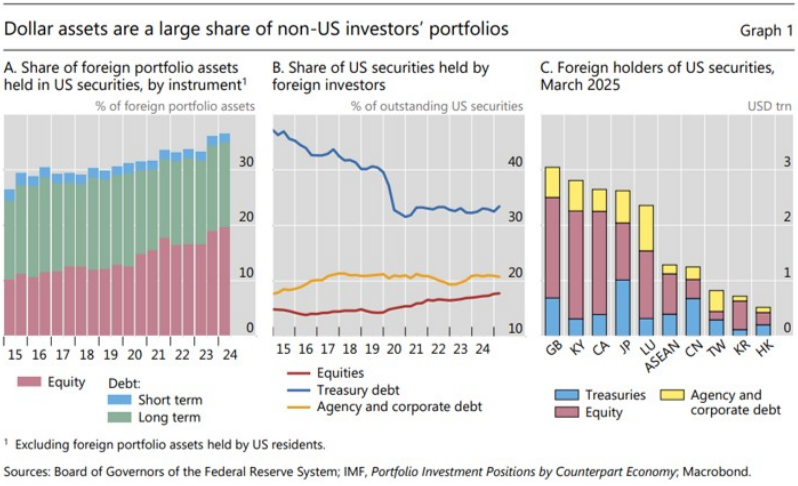
- **Setting:** Taiwanese life insurers manage close to USD 1 trillion, largely in USD bonds; hedge ratios near historic lows (BofA estimate ≈65% at end-2024) because hedging was expensive; exporters held large USD balances
- **Trigger:** broad USD weakness after the April tariff shock + speculation that a stronger TWD could be part of US–Taiwan trade talks
- **Scale:** TWD up as much as 5% intraday on 5 May – biggest intraday gain in more than three decades; more than 6% over two sessions
- **Mechanics:** exporters and insurers sell USD *at the same time* → one-sided customer order flow; the usual absorbers (central bank, dealers) step back; insurers' FX losses ≈ USD 1.2 bn in May
- **Policy response:** the Financial Supervisory Commission met insurers to assess their FX exposure

Lesson

A hedge ratio is a financial-stability variable. When a large, homogeneous investor class changes it at the same time, the resulting order flow can dominate a currency regardless of fundamentals.

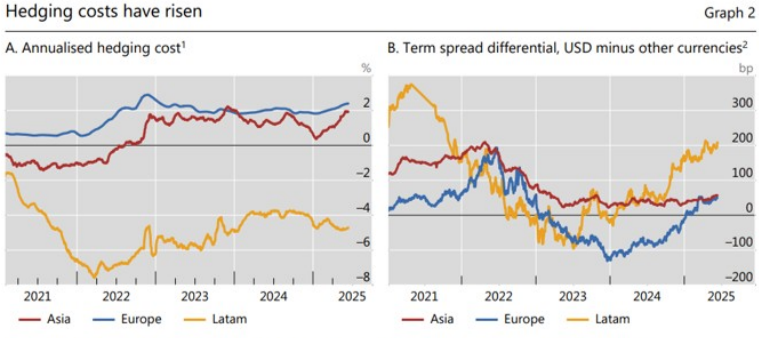
Source: Taipei Times (6 May 2025); Bloomberg; BIS Bulletin No. 105.

Hedging Activity and Non-US Investors



Source: Shin, H. S., Wooldridge, P. and Xia, D. (2025), "US dollar's slide in April 2025: the role of FX hedging", BIS Bulletin No. 105.

Hedging Costs



Source: Shin, H. S., Wooldridge, P. and Xia, D. (2025), "US dollar's slide in April 2025: the role of FX hedging", BIS Bulletin No. 105.

Hedging as a Source of Order Flow (Link to Microstructure)

- Hedging requirements of institutional investors (pension funds, insurance companies, asset managers) = significant part of customer order flow for dealers
- In Evans-Lyons model: hedging = classic example of „portfolio shifts“ and liquidity/hedging demands
- Changes in hedge ratio → large, predictable flows → affect cumulative order flow → exchange rate change
- Explains part of the „exchange rate disconnect puzzle“: rate moves even without changes in fundamentals (due to technical hedging flows)
- Policy implications: central banks today actively monitor hedge ratio changes as a leading indicator of volatility

Algorithmic vs. High-Frequency Trading (HFT)

Definition

Algorithmic Trading (AT) = automated generation and execution of orders according to a pre-defined algorithm (objective function + constraints)

Definition

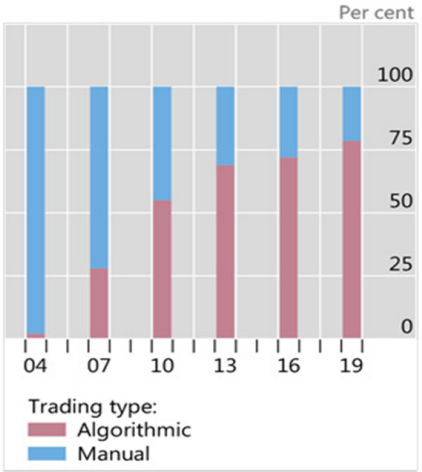
High-Frequency Trading (HFT) = special subset of AT with extreme speed:

- Thousands to millions of orders per second
- Position holding time usually under 5 seconds (often milliseconds)
- Very small volumes per trade, low margins
- Nearly zero open positions at end of day

- BIS 2025: hedge funds and Proprietary Trading Firms (PTFs, incl. HFT firms) together = **8%** of global turnover (up from 7% in 2022)
- Estimates in liquid pairs (EUR/USD, GBP/USD, USD/JPY): 40–60% of electronic spot volume
- Platforms: EBS, Reuters, multibank systems (since 2004)

Growth of Algorithmic (AT) and High-Frequency Trading (HFT)

- Creation of special trading platforms in major trading systems (2002 CME, 2004 EBS)



Technical Requirements for HFT and Network Latency

Technical requirements

- **Ultra-low latency** – under 1 ms (ideally microseconds)
- **Co-location** – servers in the data centres of exchanges/platforms (LD4 London, NY4 New York, TY3 Tokyo)
- Microwave and fibre-optic links (lower latency than the public internet)
- FPGA / ASIC chips for hardware data processing (faster than CPUs)
- Direct data feeds (market-by-order, not just market-by-price)

Network latency = delay between an event (price change, execution or cancellation of a large trade) and receiving the information

- Risks: quoting off-market (stale quotes), being arbitraged, wrong speculative/hedging decisions
- Sources: distance between nodes, data volume, number of elements the data passes through, processing capacity, trading protocols, counterparty credit checks, age of infrastructure
- Regional order books must be synchronised (book synchronisation)
- Latency is a random variable that is hard to predict $\Rightarrow$ time and price priority of orders is hard to guarantee

Latency creates adverse-selection risk for slower quoters and arbitrage opportunities for faster traders.

Main HFT Strategies in the FX Market

1. **Liquidity providing / market making**
   - Passive provision of bid/ask quotes; profit from the spread + platform rebates
   - Auto-hedging: new positions hedged immediately
   - Most common strategy (majority of HFT in FX)
2. **Arbitrage**
   - Triangular (EUR/USD + USD/JPY + EUR/JPY)
   - Cross-venue (EBS vs. LSEG Matching vs. multi-bank platforms)
   - Latency arbitrage (seeing price changes before others)
3. **Statistical / quantitative**
   - Mean reversion and pairs trading (EUR/USD vs. GBP/USD)
   - Momentum on tick data; correlation trading (commodities, equities, yields)
   - Order-book imbalance detection $\rightarrow$ short-term direction
4. **News / event-driven**
   - NLP on news feeds (keywords, sentiment)
   - Reaction to data releases (NFP, CPI, central bank decisions)

Order anticipation: trading ahead of detected large orders; its abusive forms (quote stuffing, layering, spoofing) are prohibited market manipulation (EU MAR / MiFID II, US rules)

Common features

Best environment: high liquidity + low volatility. When volatility jumps most HFTs withdraw (liquidity dry-up), and correlated strategies produce herding in a shock.

Impacts of HFT on the FX Market

Positive Impacts

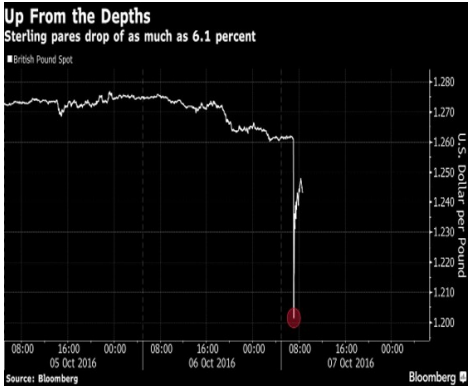
- Significant narrowing of bid-ask spreads (better liquidity)
- Faster price discovery
- Higher overall liquidity in calm periods
- Lower transaction costs for other participants

Negative Impacts

- „Flash crashes“ and sharp fluctuations (e.g., GBP 2016, JPY 2019; CHF 2015 was a policy shock that turned into a liquidity blackout)
- Liquidity withdrawal during shock (HFT becomes a liquidity taker)
- Adverse selection for classic market makers
- Increased short-term volatility and tail risk
- Procyclicality (correlation with VIX)
- Endogeneity problem of AT (HFT) – growth in AT (HFT) activity in response to rising price volatility

BIS / IMF: HFT amplifies shocks during risk appetite decline but improves normal liquidity.

GBP Flash Crash, 7 October 2016

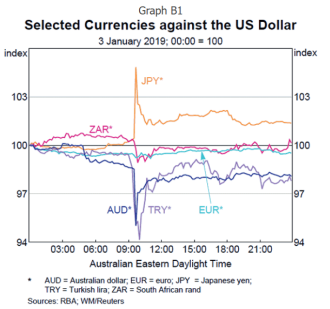


- **Timing:** ≈00:07 London time (BST) = 07:07 Singapore/Hong Kong – the "twilight zone" between the New York close and the build-up of Asian liquidity.
- **Scale:** the low depends on the venue: ≈6% on Bloomberg's composite quote (chart), ≈9% on some trading venues (BIS) – nearly 10% within about 40 seconds; most of the move retraced within about 10 minutes.
- **Trigger:** no single cause (BIS Markets Committee, 2017): time of day, sizeable sterling sales including options-related hedging, execution of stop-losses. A news headline on a "hard Brexit" (F. Hollande) was widely cited at the time, but was not established as the cause.
- **Mechanics:** order-flow imbalance + market makers stepping back; an automatic pause in sterling futures removed a price reference for cash-market liquidity providers (Bank of England analysis).
- **Lesson:** algorithms can amplify even marginal news into systemic shocks if they hit an "empty" order book.

Source: Bloomberg, 7 October 2016 (chart); BIS Markets Committee (2017), "The sterling 'flash event' of 7 October 2016"; Bank Underground (8 March 2018).

Case Study: Yen Flash Event, 3 January 2019

- **Timing:** ≈07:30 Tokyo time (9:30 Sydney); Japanese banks – key yen market makers – closed for the New Year holiday
- **Context:** Apple's revenue warning an hour earlier → weaker risk sentiment; Japanese retail investors held large long positions in high-yielding currencies against the yen
- **Scale:** yen $+3\%$ against USD in about 30 seconds; AUD/JPY -7%; AUD/USD briefly at a 10-year low (0.6715); bid–ask spreads widened from ≈2 pips to 100–300 pips
- **Mechanics:** thin liquidity → automatic stop-outs of retail margin accounts → liquidity-providing algorithms switched off, momentum algorithms amplified
- **Recovery:** about half of the move retraced within minutes; spreads back to normal within an hour



Lesson

Flash events need two ingredients: crowded, mechanically managed positions (stop-losses, margin rules) and a moment when the capacity of the market to absorb order flow (γ in the Evans–Lyons model) is close to zero.

Source: Reserve Bank of Australia, Statement on Monetary Policy, February 2019, Box B.

Case Study: SNB Floor Removal, 15 January 2015

- **Background:** from September 2011 the SNB defended a minimum rate of **1.20 EUR/CHF** by buying euros without limit; the market treated it as „bulletproof“ – buy orders at 1.2001, stop-losses just below 1.20.
- **10:30 CET:** SNB unexpectedly announces immediate removal of the exchange rate ceiling.
- **Market Reaction:** Within minutes, the franc strengthened by over **30%** (short-term reaching 0.85 EUR/CHF).
- **Microstructural Collapse:**
  - Dealers and algorithms stopped quoting (bid shutdown).
  - *Liquidity Blackout:* No one in the market wanted to buy EUR for CHF.
  - *Gap Risk:* Price didn't jump from 1.20 to 1.19 smoothly but „gapped“ directly to 1.00 without execution possibility in between.

Microstructure Failure: The Slippage and Stop-loss Problem

- **Illusion of Protection:** Many retail traders had stop-losses set at 1.1990. Due to absence of liquidity, these orders were executed at levels around 1.05 or 0.95.
- **Negative Balance:** Slippage was so massive that clients lost their entire deposits and ended up in deep debt to brokers.
- **Impact on Brokers and Funds:**
  - *Alpari UK:* Declared insolvency within 24 hours.
  - *FXCM:* Had to take a USD 300 million rescue loan to survive.
  - Large banks (Citigroup, Deutsche Bank) reported losses in the hundreds of millions of USD.
  - *Everest Capital Global Fund* ($\approx$USD 830 m), positioned for a weaker franc, lost almost all its capital and closed.

Case Study (Contrast): ČNB Exchange Rate Commitment, 2013–2017

Design and exit

- 7 Nov 2013: ČNB commits to prevent koruna appreciation below CZK 27/EUR (one-sided floor)
- Earliest exit date communicated in advance (forward guidance)
- Interventions accelerated before the exit: cumulative $\approx$ EUR 75 bn by March 2017
- Market estimates of long-koruna speculative positions: EUR 50–60 bn
- Exit at an extraordinary Bank Board meeting, 6 April 2017

Why no SNB-style gap?

- Exit *expected*: the market was **long** CZK, so the exit triggered profit-taking – two-sided order flow
- Exporters had already hedged forward
- EUR/CZK stayed just below 27 for weeks
- SNB 2015: surprise + one-sided positioning (stops below 1.20) + no absorber $\Rightarrow$ gap

Lesson

Exiting a peg removes the central bank as the absorber of order flow. What follows depends on how the market is positioned and whether the exit was anticipated.

Hedge Funds – Definition and Characteristics

- Limited-regulation collective investment fund (typically Cayman, Delaware, Luxembourg)
- Goal: Absolute return (positive return) regardless of market development – benchmark independent
- Highly active, flexible strategies using leverage (3–10×) and derivatives
- Investors: High-net-worth individuals (HNW), institutions (pension funds, foundations, family offices) + manager's own capital
- **Global hedge fund capital at end of 2025: Record USD 5.15 trillion** (HFR)
- Record increase in 2025: +USD 642.8 billion (USD 527 bn performance + USD 116 bn net inflows – the strongest inflows since 2007)
- Macro funds (the most FX-relevant strategy): USD 787 billion
- In the FX market: hedge funds and PTFs = 8% of turnover; the broader „other financial institutions" category = 50% (BIS 2025)

Source: HFR Global Hedge Fund Industry Report, 22 Jan 2026

Hedge Fund Strategies through an FX Lens

Strategy	Typical FX position	Order-flow footprint	When it hurts
Global macro (discretionary)	Directional bets on policy divergence, pegs, regime shifts	Lumpy, concentrated, often informed	Policy surprises against the position (SNB 2015)
Systematic carry	Long high-yield, short low-yield currencies	Slow, persistent build-up	Risk-off, funding-currency rallies (1998, 2008, 2024)
Trend / CTA	Follows past returns, positions scaled by volatility	Procyclical, mechanical, predictable	Trend reversals, volatility spikes
Volatility / options	Short or long implied volatility and skew	Delta-hedging flows of option dealers	Volatility spikes, gap moves
Relative value / basis	Hedged spreads (CIP basis, cross rates)	Small net, large gross positions	Funding and balance-sheet shocks
Event-driven	Central-bank meetings, elections, peg exits	Concentrated around events	Binary outcomes, gap risk

Key question for financial stability: are positions **crowded** and **mechanically managed**? If so, the exit becomes correlated, one-sided order flow.

Access: funds trade through prime brokers (give-up to executing dealers), using spot, forwards, NDFs for restricted currencies (INR, KRW, TWD, BRL), options and CME futures – the investor base, fees, prime brokerage and leverage are covered in the lecture *Financial System Integration and Shock Transmission*.

Global Macro: Betting on Policy Regimes

- Discretionary, top-down: look for **inconsistencies** between policy and fundamentals (overvalued pegs, unsustainable divergence, fiscal–monetary conflict)
- Why shorting a defended currency is attractive – an asymmetric („one-way“) bet:

$$\mathbb{E}[\text{payoff}] \approx p \cdot d \; - \; (1-p) \cdot (i - i^*)\,\tau$$

  $p =$ probability that the peg breaks, $d =$ expected devaluation, $(i - i^*)\tau =$ carry cost of the short position over horizon τ
- If the peg holds, the loss is roughly the interest differential; if it breaks, the gain is large $\Rightarrow$ even a small p justifies a big position
- Instruments: forwards, NDFs, out-of-the-money options (convexity at low cost)
- Self-fulfilling element: large short positions raise the cost of defence $\Rightarrow$ p rises (second-generation crisis models, Obstfeld 1996)
- Defence tools: reserves, rate hikes, capital controls – each with a domestic cost

Macro hedge funds manage $\approx$ USD 0.8 trillion (HFR, end-2025).

Case Study: Black Wednesday, 16 September 1992

- **Setting:** GBP in the ERM (central rate DM 2.95, floor $\approx$ DM 2.78); UK in recession while the Bundesbank kept rates high after reunification $\Rightarrow$ inconsistent policy mix
- **Positioning:** macro funds (Soros' Quantum Fund $\approx$ USD 10 bn short GBP) and banks build large short-sterling positions – a one-way bet
- **Defence:** the Bank of England buys sterling on a massive scale; base rate raised from 10% to 12%, with 15% announced the same day
- **Outcome:** UK suspends ERM membership that evening; sterling falls by more than 10% against the DM within weeks; Quantum's profit $\approx$ USD 1 bn; HM Treasury later put the cost at $\approx$ £3.3 bn

Lesson

A peg is a promise to absorb any order flow at a fixed price. It is only as strong as the central bank's reserves and its willingness to bear the domestic cost of defending it.

Systematic FX Strategies: Carry, Momentum, Value

Let $s_{j,t}$ = log units of currency j per USD; excess return from holding currency j:

$$rx_{j,t+1} = f_{j,t} - s_{j,t+1} \approx (i_{j,t} - i_{US,t}) - \Delta s_{j,t+1}$$

Each month, sort currencies on a signal; go long the top portfolio and short the bottom one.

	Signal	**Return driver**	**Crash profile**
Carry	Forward discount $f_j - s_j \approx i_j - i_{US}$	Compensation for global FX volatility risk (Lustig et al. 2011; Menkhoff et al. 2012a)	Negative skew; crashes in risk-off
Momentum	Past 1–12 month excess return	Under-reaction, slow-moving capital (Menkhoff et al. 2012b)	Sharp reversals at turning points
Value	Deviation of the real exchange rate from its long-run level (e.g. 5-year change)	Slow mean reversion towards PPP (Asness et al. 2013)	Long periods of under-performance

- Low correlation between the three factors $\Rightarrow$ multi-factor "currency overlay" programmes
- When many funds trade the same signals, exits become correlated – factor crowding

Trend Followers and Volatility Targeting: Mechanical Order Flow

- CTAs / managed futures trade liquid FX futures and forwards using rules, e.g. time-series momentum (Moskowitz, Ooi & Pedersen 2012)
- Typical position rule with volatility targeting:

$$\text{position}_{j,t} = \underbrace{\text{sign}\left(r_{j,\,t-12m:t}\right)}_{\text{trend signal}} \times \underbrace{\frac{\sigma^{target}}{\hat{\sigma}_{j,t}}}_{\text{risk scaling}} \times \text{capital}_t$$

- Implications for order flow:
  - $\hat{\sigma} \uparrow \Rightarrow$ positions are cut mechanically, *whatever the fundamentals* $\Rightarrow$ selling into a falling market
  - rules are similar across funds $\Rightarrow$ flows are **predictable** and can be front-run
  - when a trend reverses, many funds flip at similar thresholds $\Rightarrow$ correlated, one-sided flow
- The same logic applies to risk-parity funds and to VaR-based limits at banks

August 2024: deleveraging and margin increases amplified the initial reaction to weak US data (BIS Bulletin No. 90).

Relative Value and Volatility Strategies in FX

Volatility selling

- Implied volatility is on average above realised volatility $\Rightarrow$ selling straddles/strangles earns a **volatility risk premium** (Della Corte, Ramadorai & Sarno 2016)
- Payoff: small steady gains, rare large losses („picking up nickels in front of a steamroller“)
- Who is short gamma matters: option dealers' delta hedging can amplify moves (options-related selling in GBP, October 2016)

Relative value and event-driven

- CIP basis trades: lend dollars via FX swaps when the basis is wide – limited by balance-sheet costs (Du, Tepper & Verdelhan 2018)
- Cross-rate and triangular mispricings – today mostly arbitraged by HFT/PTFs
- Relative value between correlated currencies (NOK vs SEK, AUD vs NZD)
- Event-driven: central-bank meetings, elections, referenda (Brexit 2016), peg exits – binary outcomes with gap risk

Positive Impacts of Hedge Funds on Financial Stability

- **Liquidity Provider** – high activity, frequent portfolio changes, HFT, use of leverage
- Focus on „risky“ instruments (emerging markets, distressed debt, volatility, carry trades)
- Trading against the trend or independently of the trend $\rightarrow$ increases market integration
- Volatility absorption (searching for high volatility markets)
- High risk-adjusted return for investors

BIS and IMF: Hedge funds contribute significantly to price discovery and liquidity in normal periods.

Negative Impacts and Systemic Risks

- **Fire sales** during margin calls → transfer of problems to prime brokers and banks
- Procyclicality: Trend following, stop-loss, liquidation during risk appetite decline
- Threat to illiquid markets (EM FX, credit, structured products)
- Strategy correlation: Many funds behave similarly during shock (herding)
- Risk transfer: Client currency risk → bank credit risk (example: CHF mortgages 2015)

Example: Aug 2024 yen carry unwind – massive hedge fund positions amplified global sell-off.

Case Study: Autumn 1998 – LTCM, Deleveraging and the Yen

- **Setting:** near-zero Japanese rates made the yen the funding currency of choice; the dollar rose to ≈¥147 in August 1998 (see the USD/JPY chart in the carry-trade section)
- **Shock:** Russian default (17 August 1998) → losses at LTCM (≈USD 4.6 bn) and other leveraged funds; Fed-brokered recapitalisation of LTCM by 14 institutions (23 September)
- **Unwind:** hedge funds and other speculative accounts liquidated long-dollar positions to cut risk and cover losses in other markets
- **Scale:** 7 October: USD/JPY from ¥133.90 to ¥120.55 – the largest one-day move since 1974; low of ¥111.58 on 8 October
- **Microstructure:** direct inter-bank dealing declined; some banks quoted spreads several times wider than usual

Lesson

Funding currencies appreciate in risk-off episodes – exactly when leveraged carry traders are forced to buy them back. „Down by the elevator."

Source: Federal Reserve Bank of New York, Treasury and Federal Reserve Foreign Exchange Operations, October–December 1998.

Monitoring Crowded Positions: What Supervisors Look At

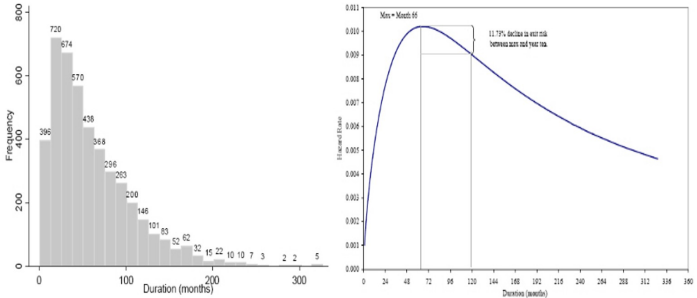
- **CFTC Commitments of Traders**: weekly net positions of "leveraged funds" in CME currency futures (e.g. net yen shorts at their largest since 2007 in early July 2024)
- **Options market**: risk reversals and implied volatility – the price of crash insurance
- **Carry indicators**: carry-to-risk ratios, interest differentials relative to implied volatility
- **Derivatives reporting**: trade repositories (EMIR in the EU), BIS Triennial and semi-annual derivatives statistics
- **Leverage and funding**: prime-brokerage and margin data, Form PF (US), AIFMD reporting (EU)
- Size is hard to measure: the BIS put yen carry before August 2024 at $\approx$ ¥40 trillion (USD 250 bn), probably biased low because of data gaps

Crowding matters more than size: exits are correlated when positions are similar and managed by similar rules.

Empirical Evidence – Returns, Correlations, and Defaults

- Average annual hedge fund return 2000–2025: 7–9% (HFR HFRI index)
- Correlation with equity market (S&P 500): 0.4–0.7 (higher than before)
- Correlation with volatility: often negative (volatility absorption)
- Attrition („default“ = liquidation or closure of the fund): 8–12% of funds annually (higher than traditional funds)
- Largest drawdowns: 2008 (–20%), 2020 (–10%), 2022 (–8%)

Hedge Fund Default Frequency and Probability of Default



Source: Grecu, A., Malkiel, B. G. and Saha, A. (2006), "Why do hedge funds stop reporting their performance?", Princeton CEPS Working Paper 124; TASS data, 1996–2004.

Empirical Evidence – Correlations by Strategy (1990–2014)

Figure 1: Correlation table comparing stocks, bonds and hedge fund sub-sectors

From Jan. 1990 to Jan. 2014		1	2	3	4	5	6	7	8
Developed Market Equities	1	1.00							
Developed Government Bonds	2	-0.01	1.00						
Event-Driven Index	3	0.61	-0.09	1.00					
Equity Hedge Index	4	0.63	-0.05	0.84	1.00				
Equity Market Neutral Index	5	0.16	0.04	0.42	0.51	1.00			
Quantitative Directional	6	0.71	-0.02	0.82	0.89	0.34	1.00		
Macro Index	7	0.34	0.35	0.51	0.55	0.34	0.55	1.00	
Relative Value Index	8	0.41	-0.05	0.77	0.68	0.41	0.57	0.34	1.00

Source: Bloomberg, HFRI, Barclays.

Source: Bloomberg, HFRI, Barclays.

- Monthly HFRI strategy indices vs developed-market equities and government bonds, Jan 1990 – Jan 2014
- **Equity beta:** quantitative directional 0.71, equity hedge 0.63, event-driven 0.61 with equities
- **Diversifiers:** equity market neutral 0.16, macro 0.34; macro is the only strategy correlated with bonds (0.35)
- **Near-duplicates:** equity hedge vs quantitative directional 0.89, event-driven vs equity hedge 0.84
- **For FX:** macro – the main FX strategy – is the least tied to equities; its risk sits in rates and currencies
- **Caveats:** full-sample averages – correlations rise in crises (2008); self-reported index data understate risk (survivorship, smoothed returns)

The Meese-Rogoff Puzzle (1983)

Key Finding

Structural fundamental-based models (Monetary model, Dornbusch overshooting model) are systematically outperformed in exchange rate forecasting over short to medium horizons by a simple **Random Walk** ($\hat{s}_{t+h} = s_t$).

- **What is a Random Walk?** Today's exchange rate is the best predictor of tomorrow's. All available information is already reflected in the current price.
- **Theoretical Pillars**: Purchasing Power Parity (PPP), Uncovered Interest Parity (UIP), and Rational Expectations.
- **Empirical Failure**: R^2 for monthly predictions typically ranges between 0–10%. Even at 1–2 year horizons, macro models add almost no predictive power.
- **Conclusion for Policymakers & Traders**: The exchange rate behaves like a "loose cannon" — largely disconnected from real economic fundamentals in the short-to-medium run.

Exchange Rate Forecasting Methods (Traditional)

1. **Naive (Random Walk)**: The best estimate of the future rate is today's rate. Simple, hard to beat.
2. **Fundamental (Long-Run)**: Uses PPP or Monetary models to determine long-term trends.
   - Example (Inflation differential): $\hat{s}_{t+h} = s_t + h(\bar{\pi} - \bar{\pi}^*)$
3. **Hybrid (Error Correction Models – ECM)**: Combines short-run dynamics with long-run fundamental relationships (cointegration):

$$\Delta s_t = \alpha(s_{t-1} - \beta' x_{t-1}) + \gamma \Delta x_t + \epsilon_t$$

   (Where α is the speed of adjustment toward the fundamental equilibrium.)

Key takeaway: These methods work reasonably well for very long horizons, but fail dramatically at policy-relevant horizons (3–24 months).

UIP and Time-Varying Risk Premia (TVRP)

Why do structural models fail? They ignore risk.
In reality, pure UIP does not hold. The correct relationship must include a risk premium (rp_t):

$$i_t - i_t^* = E_t[\Delta s_{t+1}] + \underbrace{rp_t}_{\text{Time-Varying Risk Premium}}$$

- **UIP Failure (Forward Bias Puzzle)**: rp_t is highly volatile and often dominates the interest differential. High-interest currencies tend to appreciate, not depreciate.
- **Volatility Amplification**: Shocks to risk aversion (changes in rp_t) cause sudden jumps in the exchange rate that appear unrelated to macroeconomic news.
- **Implication**: The exchange rate reacts less to the real economy and more to **sentiment, risk appetite, and financial conditions**.

Engel-West (2005): The Present Value Model

This framework reconciles the random walk behavior with the existence of fundamentals by viewing the exchange rate as an **asset price**.

Model Structure: The spot rate (s_t) is the discounted sum of expected future fundamentals (f_{t+k}):

$$s_t = (1-b)\sum_{k=0}^{\infty} b^k E_t[f_{t+k}]$$

Key Finding (Engel-West)

If fundamentals (f_t) are highly persistent (near a random walk) and the discount factor b is close to one, then the **spot rate will behave like a random walk**, even if it is fully determined by fundamentals.

- **Intuition**: Markets price in next year's expectations *today*. Good news about future fundamentals is already in the price.
- **Conclusion**: Low predictive power of fundamentals is not necessarily a flaw in the theory — it is a natural consequence of the **forward-looking nature of asset markets**.

What is FX Market Microstructure?

- Microstructure = process of price formation through trading interaction.
- We are not interested in „model equilibrium“, but:
  - Who trades?
  - In what way?
  - Who bears inventory risk?
  - How is information aggregated?
- Price is formed in the **order book**, not in a „UIP equation“.

Key question: How does private information get into the exchange rate?

RFQ vs. Order Book System

- **RFQ (Request for Quote)**
  - Typical for institutional clients,
  - Dealer quotes a tailored price.
- **Central Limit Order Book (CLOB)**
  - EBS, Reuters
  - Anonymous electronic trading
  - Time and price priority

Modern FX market = hybrid structure.

Measuring Order Flow

- Every trade has a buyer and a seller – volume alone has no direction
- Order flow signs each trade by its **initiator** (aggressor): the side that demands immediacy and crosses the spread

$$\Delta x_t = \sum_{k \in t} d_k \, q_k, \qquad d_k = \begin{cases} +1 & \text{buyer-initiated} \\ -1 & \text{seller-initiated} \end{cases}$$

- q_k = trade size (or $q_k \equiv 1$: number of trades, as in the Evans–Lyons 1996 data)
- Cumulative order flow $x_t = \sum_{\tau \le t} \Delta x_\tau$ co-moves with the *level* of the exchange rate
- Data: inter-dealer platforms (EBS, Reuters/LSEG), dealers' customer flows, CLS settlement data

Example (EUR/USD, EUR m)

Trade	Price	Initiator	$d_k q_k$
1	1.0852 (ask)	buyer	$+5$
2	1.0851 (bid)	seller	-2
3	1.0853 (ask)	buyer	$+3$
4	1.0852 (bid)	seller	-1
Volume			11
Order flow Δx			$+5$

Why Should Order Flow Move Prices? Three Channels

Channel	Mechanism	Persistence	Classic references
Inventory	Risk-averse dealer shades quotes to unload unwanted inventory	Transitory – reverses as inventory is laid off	Stoll (1978); Ho & Stoll (1981)
Information (adverse selection)	Orders of informed traders reveal private information about value; dealers update quotes	Permanent	Kyle (1985); Glosten & Milgrom (1985)
Portfolio balance	Currencies are imperfect substitutes: absorbing a shift in net supply requires a change in the risk premium	Persistent – as long as holdings stay shifted	Lyons (2001); Evans & Lyons (2002)

Kyle (1985) benchmark: $\Delta p = \lambda \cdot (\text{order flow})$ with $\lambda = \dfrac{\sigma_v}{2\sigma_u}$ – impact rises with private information (σ_v) and falls with noise trading (σ_u)

The Evans–Lyons twist

In FX there is little private information about *payoffs*. The private information is about **aggregate demand** (portfolio shifts) – dispersed across dealers and revealed only through order flow.

Evans–Lyons (2002) Portfolio Shifts Model: Idea and Setup

Key idea

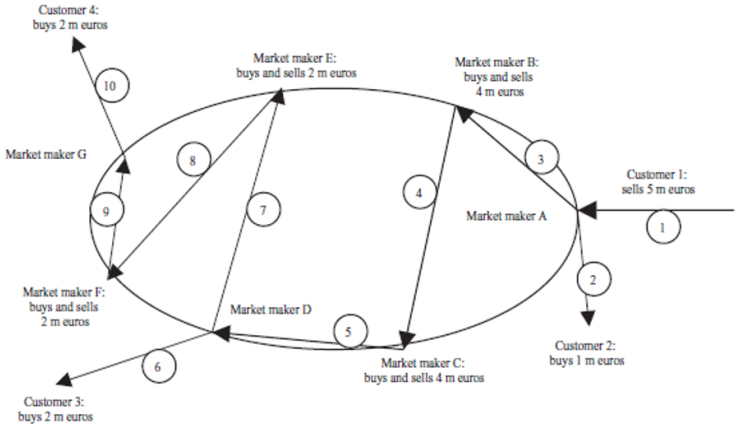
The exchange rate moves not only with public news but through a **learning process from trading**. Order flow Δx_t (buyer- minus seller-initiated trades – net pressure, not volume) carries **dispersed information** that is not yet in the price.

- Decentralised dealer market: N risk-averse dealers (CARA utility $-\exp(-\theta W)$) + a continuum of public customers
- One risky asset (foreign exchange) with payoff $R_t = \sum_{\tau \le t} \Delta R_\tau$; the increment ΔR_t (e.g. interest-rate news) is **public** at the start of day t
- Round-1 customer orders $c_{i1} \sim N(0, \Sigma_c) =$ **portfolio shifts** (hedging, rebalancing): independent across dealers and of ΔR_t, seen only by dealer i ($c_{i1} > 0$: customer buys FX)
- Aggregate shift $c_1 = \sum_i c_{i1}$ is not news about payoffs, but someone must **absorb** it by the end of the day
- Round-3 demand of the public, with aggregate risk-bearing capacity γ:

$$c_3 = \gamma \left(\mathbb{E}[\text{payoff} \mid \Omega_3] - P_{3,t} \right), \qquad \gamma = \frac{1}{\theta_{pub} \operatorname{Var}(\text{payoff})}$$

- Information is not common knowledge $\Rightarrow$ order flow acts as a signal

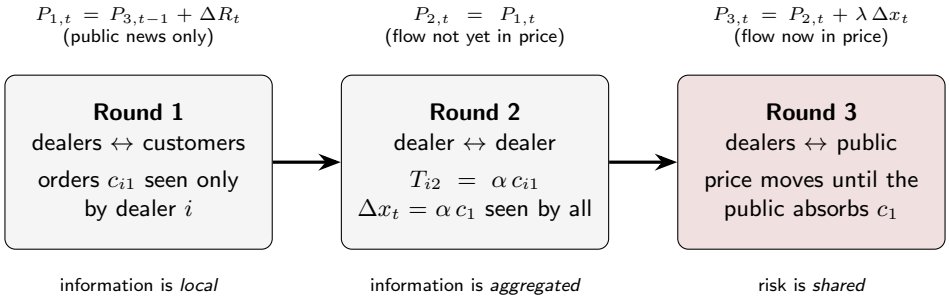
Trading Process in the FX Market



Order Flow and Cumulative Order Flow

Transaction	Initiating party	Passive party	Order flow (m €)	Cumulative order flow (m €)	Total turnover (m €)
1	Customer 1	Market maker A	-5	-5	5
2	Customer 2	Market maker A	+1	-4	6
3	Market maker A	Market maker B	-4	-8	10
4	Market maker B	Market maker C	-4	-12	14
5	Market maker C	Market maker D	-4	-16	18
6	Customer 3	Market maker D	+2	-14	20
7	Market maker D	Market maker E	-2	-16	22
8	Market maker E	Market maker F	-2	-18	24
9	Market maker F	Market maker G	-2	-20	26
10	Customer 4	Market maker G	+2	-18	28

One Trading Day in Evans–Lyons



- **Round 1:** customers trade with their own dealer – each dealer sees only its own clients' orders
- **Round 2:** „hot potato" trading – dealers pass inventory to each other; through these trades the private signals become **observable to all**
- **Round 3:** dealers move the price until the public takes over the remaining risk

Order flow is the channel through which dispersed information about the aggregate portfolio shift c_1 reaches the price.

Evans–Lyons: Rounds 1 and 2

- **Round 1**: dealers' quotes must be common (otherwise arbitrage) and can only reflect public information:
$$P_{1,t} = P_{3,t-1} + \Delta R_t$$
- **Round 2**: inter-dealer quotes are set simultaneously and independently $\Rightarrow$ they cannot yet reflect c_1: $P_{2,t} = P_{1,t}$
- Dealer i's inter-dealer trade:
$$T_{i2} = \alpha\, c_{i1}, \qquad \alpha > 0$$
– lays off the customer position and speculates on own information („hot potato")
- Inter-dealer order flow: $\Delta x_t = \sum_i T_{i2} = \alpha\, c_1$
- End of round 2: all dealers observe $\Delta x_t \Rightarrow$ infer the portfolio shift *exactly*: $c_1 = \Delta x_t / \alpha$

Evans–Lyons: Round 3 and the Price Impact λ

Simplified one-period version (Lyons 2001, ch. 7); the dynamic model in Evans & Lyons (2002) yields the same λ.

1. After rounds 1–2 dealers as a group are short what customers bought: aggregate position $-c_1$
2. Dealers are risk averse and want to end the day flat $\Rightarrow$ the public must take the other side: $c_3 = -c_1$
3. Market clearing in round 3 (normalising $P_{2,t} = \mathbb{E}[\text{payoff} \mid \text{public info}]$):

$$\gamma\,(P_{2,t} - P_{3,t}) = -c_1 \;\Rightarrow\; P_{3,t} = P_{2,t} + \frac{c_1}{\gamma} = P_{2,t} + \frac{1}{\alpha\gamma}\,\Delta x_t$$

4. Combining with round 1:

$$\boxed{\Delta P_t = \Delta R_t + \lambda\,\Delta x_t, \qquad \lambda = \frac{1}{\alpha\gamma}}$$

Intuition

Net dollar buying ($\Delta x_t > 0$) raises the price of the dollar until the public is willing to sell the dollars dealers need. The effect persists: it is a change in the risk premium needed to hold a different portfolio, not a temporary inventory effect.

Key Model Equation

$$\Delta p_t = r_t + \lambda \Delta x_t + \epsilon_t$$

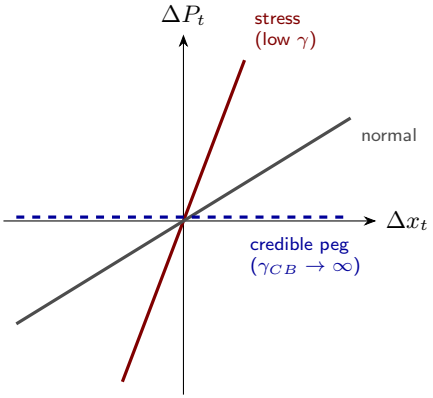
$$\lambda = \frac{1}{\alpha \cdot \gamma} > 0$$

where:

- Δp_t = change in log exchange rate
- r_t = public information (interest rates, inflation...)
- Δx_t = daily (net) inter-dealer order flow; cumulative order flow is $x_t = \sum_{\tau \leq t} \Delta x_\tau$
- α = dealers' trading intensity: inter-dealer trade per unit of customer order ($T_{i2} = \alpha c_{i1}$); it reflects dealers' risk aversion and inventory management
- γ = risk-bearing capacity of the public (elasticity of public demand)
- Per unit of *customer* flow the impact is $1/\gamma$, independent of α: what ultimately matters is who can absorb the risk

What Drives λ? Order Flow and Financial Stability

- Per unit of customer flow: $\Delta P = c_1/\gamma$, with $\gamma = 1/(\theta_{pub}\,\sigma^2)$
- **Stress**: risk aversion ↑, volatility ↑, constrained balance sheets, HFT liquidity withdrawn $\Rightarrow \gamma \downarrow \Rightarrow$ the same flow moves the price much more
- **Credible peg**: the central bank absorbs any flow at the parity, $\gamma_{CB} \to \infty \Rightarrow \lambda \to 0$ – FF/DM order flow stopped moving the rate once EMU parities were announced (Killeen, Lyons & Moore 2006)
- **Peg exit**: the absorber disappears, γ collapses $\Rightarrow$ gap (SNB 2015)
- In the data, impact is lowest when trading is most active (Berger et al. 2008)



Evans–Lyons (2002): Estimation

$$\Delta p_t = \beta_1 \, \Delta(i_t - i_t^*) + \beta_2 \, \Delta x_t + \eta_t$$

- Data: Reuters Dealing 2000-1 inter-dealer trades, 1 May – 31 August 1996 (89 trading days); Δx_t = buyer- minus seller-initiated trades (thousands); Δp_t = daily change in the log spot rate

	β_1 (interest differential)	β_2 (order flow)	R^2
DM/USD	0.52 (0.35)	2.10 (0.20)	0.64
JPY/USD	2.48 (0.92)	2.90 (0.46)	0.45

- Interest differential alone: $R^2 < 0.01$ – the „disconnect“ in miniature
- 1,000 net dollar purchases $\Rightarrow$ DM price of the dollar $+2.1\%$; with an average trade of USD 3.9 m: **USD 1 bn of net purchases** $\Rightarrow \approx +0.5\%$

Empirical Results and Implications (DM/USD 1996)

- Contemporaneous fit: order flow explains 45–64 % of daily changes (DM/USD, JPY/USD, 1996); on EBS data 1999–2004 ≈50% daily, falling to 20–30% at monthly horizons (Berger et al. 2008)
- Out-of-sample: forecasts based on customer order flow beat the random walk at horizons up to about one month (Evans & Lyons 2005, „Meese–Rogoff Redux“)
- Evans & Lyons (2012): order flow predicts future fundamentals (output, money growth, inflation) better than the spot rate does
- $\lambda \approx 0.5\%$ (≈54 bps) per 1 billion USD net purchase (DM/USD 1996); ≈40 bps per billion on EBS daily data 1999–2004
- Roughly 80% of the impact does not reverse (Evans & Lyons 2001, hourly DM/USD)
- Implications: the exchange rate adjusts gradually in real-time, not instantaneously

What Order-Flow Regressions Do – and Do Not – Show

- **Contemporaneous fit $\neq$ forecasting**: the high R^2 uses same-day flow; forecasting requires flow known *before* the price move
- **Reverse causality / feedback trading**: prices can drive flows (stop-losses, momentum rules); Danielsson & Love (2006) model this feedback explicitly
- **Sample and coverage**: four months in 1996, one inter-dealer platform; today's market is far larger and more fragmented (internalisation, many venues)
- **Transitory vs permanent**: inventory effects reverse, information and portfolio-balance effects persist – the slope declines at longer horizons (Berger et al. 2008)
- **State dependence**: λ is not a structural constant – it rises in stress and falls in deep markets
- **Order flow is not exogenous**: it aggregates hedging, carry, CTA and central-bank flows – the economics is in *who* trades and *why*

Whose Order Flow Carries Information?

- Inter-dealer flow aggregates customer flow – the information originates with **end users**
- Customer groups differ sharply in forecasting power, trading style and risk exposure (Menkhoff, Sarno, Schmeling & Schrimpf 2016; flows of a large dealer across many currencies)
- Non-financial corporates tend to act as **liquidity providers**, financial customers as liquidity takers (Bjønnes, Rime & Solheim 2005)
- Leveraged investors: their flows reflect risk appetite and positioning and can switch sign abruptly in deleveraging
- Central banks: interventions are order flow too – even secret interventions work through the same price-impact channel (Evans & Lyons 2001)

Exercise: How Large Is a Hedging Flow?

- Investors hold USD 500 bn of unhedged US bonds and raise their hedge ratio by 10 pp: forward/swap sales of USD 50 bn, which dealers offset by selling USD spot
- Naive impact with $\lambda \approx 0.4\%$ per USD 1 bn (EBS estimate, 1999–2004): $50 \times 0.4\% = 20\%$

Discussion: why are actual moves far smaller in EUR/USD – but sometimes not in TWD?

- Today's market is much deeper (USD 9.6 trillion a day) $\Rightarrow$ λ per billion is smaller
- Flows are spread over days and weeks and meet offsetting flows (exporters, central banks, other investors)
- λ is specific to the currency and the state: thin markets, one-sided positioning and absent absorbers (TWD, May 2025) raise it
- What moves the price is the *unexpected* part of the flow – anticipated hedging is partly priced in advance

Case Study: The Central Bank as Order Flow – Japan 2022–2026

Operations	USD/JPY	Yen bought
Sep–Oct 2022 (3 days)	145–152	¥9.2 tn
29 Apr & 1 May 2024	≈160	¥9.8 tn
11–12 Jul 2024	≈161	¥5.5 tn
28 Apr – 27 May 2026	>160	¥11.7 tn (record)

Source: Japanese Ministry of Finance; media reports.

- MoF decides, BoJ executes; amounts published monthly
- In the Evans–Lyons framework intervention is a large signed order flow I_t with price impact λI_t, even if secret
- Larger effect when it **surprises** (Tokyo holidays, right after US data), when positioning is crowded, and when it signals future policy
- Limits: without a narrowing rate differential the effect fades – USD/JPY was back near 159 within weeks of the 2026 operation

Lesson

Intervention buys time and punishes one-way bets, but it does not remove the carry incentive. In 2024, the July interventions, the BoJ hike and weak US data preceded the August unwind.

The Global Financial Cycle (Rey, 2013)

Core Hypothesis

There is a single global financial cycle in capital flows, asset prices, and credit growth, primarily driven by US monetary policy and global risk appetite.

- **Key Indicator**: VIX Index ("Fear Index") — inversely correlated with the global factor.
- **Risk-On Phase (Low VIX)**:
  - Search for yield, leverage expansion.
  - Massive capital inflows to Emerging Markets via carry trades.
- **Risk-Off Phase (High VIX)**:
  - Deleveraging and flight to quality (mainly into USD assets).
  - Sharp USD appreciation and EM currency depreciation.

Implication: National monetary policies are less independent than textbooks suggest.

Shock Transmission and Amplification Mechanisms

The Financial Stability Spiral:

$$\text{Risk Shock} \rightarrow \text{Leverage Adjustment} \rightarrow \text{Order Flow (EM Sell-off)}$$
$$\rightarrow \text{FX Depreciation} \rightarrow \text{Balance Sheet Shock}$$

- **Currency Mismatch**: Many EM firms and banks have domestic-currency revenues but USD-denominated debt.
- **Balance Sheet Effect**: Local currency depreciation raises the real burden of dollar debt $\rightarrow$ credit tightening $\rightarrow$ more selling.
- **Consequence**: In emerging markets, the exchange rate does **not** act as a shock absorber — it becomes a **shock amplifier**.

USD as a Global Funding Currency

- **Dominant Currency Paradigm (Gopinath)**: The USD is the "universal unit" of global trade and finance, not just the US currency.
- **Dollar Debt**: On-balance-sheet USD debt of non-banks outside the United States $\approx$ **$13 trillion** (BIS data).
- **Hidden Dollar Debt**: non-banks outside the US owe a further $\approx$ **$26 trillion** through FX swaps, forwards and currency swaps – about twice the on-balance-sheet amount; across all counterparties, outstanding USD obligations from these contracts exceed $80 trillion (end-June 2022; Borio, McCauley & McGuire 2022)
- **FX Swaps**: The main tool non-US banks use to obtain dollar liquidity (often representing hidden dollar debt).

Key Implication for Stability

A strong dollar acts as a **contractionary global monetary policy**. When the USD appreciates, global financial conditions tighten, leading to a decline in investment and trade even outside the US.

Synthesis: Solving the "Exchange Rate Disconnect" Puzzle

- **Macro vs. Micro**: Standard models fail because they ignore the market's learning process. Microstructure (Evans-Lyons) explains why rates don't react to fundamentals immediately—they react to **order flow**, which carries private information.
- **Link to Stability**: If the exchange rate moves without a change in fundamentals (due to technical flows), it can create false signals for the real economy.
- **Hot Potato Effect**: In the inter-dealer segment, risk is passed around like a "hot potato" until absorbed by the client sector. In times of stress, this process leads to **extreme volatility** even in the absence of external news.

Case Studies through the Lens of the Model

Episode	Order-flow shock	Absorber (failed / stepped in)	Outcome
GBP 1992	Macro funds and banks sell GBP	BoE reserves and rate hikes exhausted	Peg breaks
JPY 1998	Carry unwind: forced yen buying	Inter-bank dealers withdraw	USD/JPY -10% in a day
CHF 2015	Stop-losses just below 1.20	SNB stops absorbing	Gap of $\approx$30%
GBP 2016	Options hedging, stop-losses	Market makers at a thin hour	-9% in seconds
CZK 2017	Long-koruna positions unwound	Not needed: two-sided flow	Orderly exit
JPY 2019	Retail stop-outs	Japanese banks on holiday	AUD/JPY -7%
USD 2020*	Dash for cash, swap demand	Fed swap lines step in	Basis narrows
GBP 2022*	LDI collateral calls	BoE gilt purchases	Record GBP low
JPY 2024	Carry unwind, deleveraging	Volatility-scaled investors cut	Yen $+12\%$, VIX spike
TWD 2025	Insurers and exporters sell USD	Central bank steps back	$+5\%$ intraday
JPY 2026	MoF buys ¥11.7 tn	Central bank as absorber	Temporary relief

Common thread: prices jump when one-sided order flow meets a market whose absorbing capacity (γ) has collapsed.

* Case study in the lecture *Financial System Integration and Shock Transmission*.

Procyclicality and Systemic Risks

Key Factors in Shock Amplification

The combination of **Carry Trades + HFT + Hedge Funds** creates a powerful procyclical force.

- **Liquidity Dry-up**: HFTs provide liquidity during calm periods but vanish during shocks, leading to "liquidity black holes" (e.g., SNB 2015).
- **Negative Skewness & Crash Risk**: Systemic underestimation of risk in carry trades leads to asymmetry—currencies appreciate slowly but collapse instantly.
- **Margin Calls**: High leverage forces funds to sell off even healthy assets, turning local shocks into global crises.

References (1/2)

- Aquilina, M., Lombardi, M. J., Schrimpf, A., & Sushko, V. (2024). The market turbulence and carry trade unwind of August 2024. *BIS Bulletin* No. 90.
- Asness, C., Moskowitz, T., & Pedersen, L. H. (2013). Value and momentum everywhere. *Journal of Finance*, 68(3), 929–985.
- Bank for International Settlements (2025). *Triennial Central Bank Survey: OTC foreign exchange turnover in April 2025*.
- Bank of England, Bank Underground (2018). The 2016 sterling flash episode.
- Berger, D. W., Chaboud, A. P., Chernenko, S. V., Howorka, E., & Wright, J. H. (2008). Order flow and exchange rate dynamics in electronic brokerage system data. *Journal of International Economics*, 75(1), 93–109.
- BIS Markets Committee (2017). The sterling 'flash event' of 7 October 2016.
- Bjønnes, G. H., Rime, D., & Solheim, H. O. A. (2005). Liquidity provision in the overnight foreign exchange market. *Journal of International Money and Finance*, 24(2), 175–196.
- Borio, C., McCauley, R., & McGuire, P. (2022). Dollar debt in FX swaps and forwards: huge, missing and growing. *BIS Quarterly Review*, December.
- Brunnermeier, M. K., Nagel, S., & Pedersen, L. H. (2009). Carry trades and currency crashes. *NBER Macroeconomics Annual 2008*, 23, 313–347.
- Chaboud, A., Rime, D., & Sushko, V. (2023). The foreign exchange market. *BIS Working Papers* No. 1094.
- Czech National Bank. A look back at the discontinued exchange rate commitment. *Inflation Report*, box.
- Danielsson, J., & Love, R. (2006). Feedback trading. *International Journal of Finance & Economics*, 11(1), 35–53.
- Della Corte, P., Ramadorai, T., & Sarno, L. (2016). Volatility risk premia and exchange rate predictability. *Journal of Financial Economics*, 120(1), 21–40.
- Du, W., Tepper, A., & Verdelhan, A. (2018). Deviations from covered interest rate parity. *Journal of Finance*, 73(3), 915–957.
- Engel, C., & West, K. D. (2005). Exchange rates and fundamentals. *Journal of Political Economy*, 113(3), 485–517.
- Evans, M. D. D., & Lyons, R. K. (2001). Portfolio balance, price impact, and secret intervention. NBER Working Paper 8356.
- Evans, M. D. D., & Lyons, R. K. (2002). Order flow and exchange rate dynamics. *Journal of Political Economy*, 110(1), 170–180.
- Evans, M. D. D., & Lyons, R. K. (2005). Meese-Rogoff redux: Micro-based exchange-rate forecasting. *American Economic Review*, 95(2), 405–414.
- Evans, M. D. D., & Lyons, R. K. (2012). Exchange rate fundamentals and order flow. *Quarterly Journal of Finance*, 2(4).
- Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319–338.
- Federal Reserve Bank of New York (1999). Treasury and Federal Reserve foreign exchange operations, October–December 1998.
- Galati, G., Heath, A., & McGuire, P. (2007). Evidence of carry trade activity. *BIS Quarterly Review*, September.

References (2/2)

- Geng, N., Scutaru, T., & Wiegand, J. (2018). Carry trade vs. deposit-driven euroization. IMF Working Paper 18/58.
- Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.
- Gopinath, G., Boz, E., Casas, C., Díez, F. J., Gourinchas, P.-O., & Plagborg-Møller, M. (2020). Dominant currency paradigm. *American Economic Review*, 110(3), 677–719.
- Grecu, A., Malkiel, B. G., & Saha, A. (2006). Why do hedge funds stop reporting their performance? Princeton CEPS Working Paper 124.
- HFR (2026). *Global Hedge Fund Industry Report*, 22 January 2026.
- Ho, T., & Stoll, H. R. (1981). Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1), 47–73.
- IMF (2025). *Global Financial Stability Report*, October, Chapter 2: Risk and resilience in the global foreign exchange market.
- Killeen, W. P., Lyons, R. K., & Moore, M. J. (2006). Fixed versus flexible: Lessons from EMS order flow. *Journal of International Money and Finance*, 25(4), 551–579.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- Lustig, H., Roussanov, N., & Verdelhan, A. (2011). Common risk factors in currency markets. *Review of Financial Studies*, 24(11), 3731–3777.
- Lyons, R. K. (2001). *The Microstructure Approach to Exchange Rates*. MIT Press.
- Meese, R., & Rogoff, K. (1983). Empirical exchange rate models of the seventies: Do they fit out of sample? *Journal of International Economics*, 14(1–2), 3–24.
- Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2012a). Carry trades and global foreign exchange volatility. *Journal of Finance*, 67(2), 681–718.
- Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2012b). Currency momentum strategies. *Journal of Financial Economics*, 106(3), 660–684.
- Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2016). Information flows in foreign exchange markets: Dissecting customer currency trades. *Journal of Finance*, 71(2), 601–634.
- Moskowitz, T., Ooi, Y. H., & Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Obstfeld, M. (1996). Models of currency crises with self-fulfilling features. *European Economic Review*, 40(3–5), 1037–1047.
- Reserve Bank of Australia (2019). Box B: The recent Japanese yen flash event. *Statement on Monetary Policy*, February.
- Rey, H. (2013). Dilemma not trilemma: The global financial cycle and monetary policy independence. *Jackson Hole Economic Policy Symposium*, Federal Reserve Bank of Kansas City.
- Schrimpf, A., & Sushko, V. (2019). FX trade execution: complex and highly fragmented. *BIS Quarterly Review*, December.
- Shin, H. S., Wooldridge, P., & Xia, D. (2025). US dollar's slide in April 2025: The role of FX hedging. *BIS Bulletin* No. 105.
- Stoll, H. R. (1978). The supply of dealer services in securities markets. *Journal of Finance*, 33(4), 1133–1151.